

# Hess's Law and Enthalpies of Formation

Combining equations and calculating  $\Delta H$  from formation data

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

**STANDARD ENTHALPIES OF FORMATION,  $\Delta H_f^\circ$  (KJ/MOL, 25 °C)**

$\text{CO}_2(\text{g}) -393.5 \cdot \text{CO}(\text{g}) -110.5 \cdot \text{H}_2\text{O}(\text{l}) -285.8 \cdot \text{H}_2\text{O}(\text{g}) -241.8 \cdot \text{CH}_4(\text{g}) -74.6$

$\text{C}_2\text{H}_4(\text{g}) +52.4 \cdot \text{C}_2\text{H}_6(\text{g}) -84.0 \cdot \text{C}_3\text{H}_8(\text{g}) -103.8 \cdot \text{C}_2\text{H}_5\text{OH}(\text{l}) -277.6 \cdot \text{C}_6\text{H}_{12}\text{O}_6(\text{s}) -1273.3$

$\text{NH}_3(\text{g}) -45.9 \cdot \text{NO}(\text{g}) +91.3 \cdot \text{CaCO}_3(\text{s}) -1207.6 \cdot \text{CaO}(\text{s}) -634.9 \cdot \text{Fe}_2\text{O}_3(\text{s}) -824.2 \cdot \text{Al}_2\text{O}_3(\text{s}) -1675.7$

Elements in their standard states:  $0 \cdot \text{M}(\text{C}_6\text{H}_{12}\text{O}_6) = 180.16 \text{ g/mol}$

$\Delta H^\circ = \sum n\Delta H_f^\circ(\text{products}) - \sum n\Delta H_f^\circ(\text{reactants})$ . Show units in every step.

## A. Understanding

1. State Hess's law and explain why it is true.

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2. Which of these have  $\Delta H_f^\circ = 0$ ?  $\text{O}_2(\text{g})$ ,  $\text{O}_3(\text{g})$ ,  $\text{N}_2(\text{g})$ ,  $\text{H}_2\text{O}(\text{l})$ ,  $\text{Fe}(\text{s})$ ,  $\text{C}(\text{graphite})$ ,  $\text{CO}_2(\text{g})$

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## B. Combining equations

3. Given:  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g})$ ,  $\Delta H = +182.6 \text{ kJ}$ , and  $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ ,  $\Delta H = -116.2 \text{ kJ}$ . (a) Find  $\Delta H$  for  $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ . (b) Hence find  $\Delta H_f^\circ$  of  $\text{NO}_2(\text{g})$ .

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### C. Using enthalpies of formation

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4. Calculate  $\Delta H^\circ$  for: (a)  $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_6(\text{g})$  (b)  $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$  (c)  $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \longrightarrow \text{Al}_2\text{O}_3(\text{s}) + 2\text{Fe}(\text{s})$  (the thermite reaction)

5. Calculate  $\Delta H^\circ$  for the combustion of propane:  $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$

6. Calculate  $\Delta H^\circ$  for  $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ , the first step in making nitric acid.

7. Use the table to find  $\Delta H^\circ$  for  $\text{H}_2\text{O}(\text{l}) \longrightarrow \text{H}_2\text{O}(\text{g})$ . What does this value represent?

### D. Challenge

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8. The combustion of ethanol,  $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \longrightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$ , has  $\Delta H^\circ = -1366.8\text{ kJ}$ . Use this and the table (ignoring the ethanol value) to calculate  $\Delta H_f^\circ$  of ethanol.

9. Calculate  $\Delta H^\circ$  for the combustion of glucose,  $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 6 \text{O}_2(\text{g}) \longrightarrow 6 \text{CO}_2(\text{g}) + 6 \text{H}_2\text{O}(\text{l})$ , and the energy released when 10.0 g of glucose is oxidized in the body.